

Regulating Radical Flux in Frontal Polymerization

Published as part of ACS Macro Letters special issue "ACS Macro Letters 15th Anniversary".

Darya A. Ivannikava, Audrey N. McMullin, Kevin A. Stewart, and Jacob J. Lessard*



Cite This: <https://doi.org/10.1021/acsmacrolett.6c00114>



Read Online

ACCESS |



Metrics & More

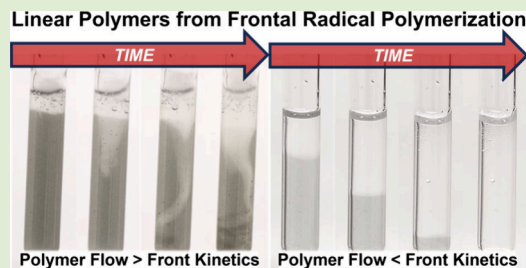


Article Recommendations



Supporting Information

ABSTRACT: In this study, we investigate the temperature dependence of radical flux in frontal radical polymerization (FRaP) for the generation of linear polymers. Linear polymers derived from FRaP under ambient pressure have historically been challenging, as the kinetics of polymerization have been outcompeted by polymer flow. Here, we demonstrate that tuning the initiator activation temperature unlocks linear polymer formation, enabling solution-state polymer characterization and providing deeper insight into the fundamental thermo-kinetic processes governing frontal polymerization. Furthermore, we introduce a multi-initiator formulation that enables sequential radical generation, mitigating premature termination while affording control over the radical flux and thermal profile of the propagating front. Collectively, these results highlight initiator selection as an important parameter governing radical flux, front stability, and the thermal characteristics of FRaP systems.



Bulk polymerization embodies the most elemental version of macromolecular synthesis: the direct conversion of monomers into polymers without dilution or additives.¹ By eliminating solvents, bulk processes channel the full chemical potential into polymer growth, maximizing conversion while minimizing impurities and downstream processing.^{2–4} In reactive manufacturing, this efficiency extends beyond synthesis—reactivity itself becomes the processing tool, allowing material shaping and curing to be dictated by polymerization kinetics and thermodynamics.

Yet, neat conventional radical polymerizations are often avoided because their high exothermicity can trigger hazardous, runaway reactions.^{5,6} As polymer chains rapidly reach high molecular weight, entanglement suppresses chain-end mobility and inhibits bimolecular termination.⁷ With termination suppressed, propagation continues unhindered, generating additional heat in a positive feedback loop. This autoaccelerative phenomenon, known as the Trommsdorff–Norris effect, often leads to uncontrolled reaction acceleration.^{8–10}

Although typically detrimental, autoacceleration can be reimagined as an asset. In carefully designed systems, suppressed termination and accelerated propagation drive rapid, localized conversion without additional external input.¹¹ Frontal polymerization, specifically frontal radical polymerization (FRaP), embodies this reframing.^{12–33} By coupling polymerization enthalpy with thermal transport, FRaP converts bulk reactivity into a self-propagating reaction front, where solvent-free formulations are not liabilities but requirements, concentrating heat to sustain the cascading reaction front. Autoacceleration sharpens this process, confining conversion to a narrow spatial domain. Thus, the very instability that limits bulk

radical polymerization becomes the driving force for scalable, energy-efficient, and structurally precise manufacturing.

In contrast, conventional solution-state radical polymerizations are often modeled using a pseudo-first-order rate approximation, assuming constant initiator decomposition and radical concentration under isothermal conditions. The rate can be simplified to $\text{rate} = k[\text{R}^\bullet][\text{M}] = k_{\text{app}}[\text{M}]$, with k_{app} encapsulating a time-invariant radical flux.^{34,35} In dilute, isothermal systems, the modest heat release from monomer conversion minimally perturbs this framework.

In FRaP, these assumptions collapse. Polymerization advances through a self-propagating thermal front where temperature rises rapidly, often surpassing initiator decomposition onset within seconds. Radical generation becomes strongly time- and position-dependent, and radical flux varies dynamically across the front.^{24,27,36} Under such nonisothermal, spatially heterogeneous conditions, the simplified rate law loses meaning, and initiator behavior must be understood in terms of its transient radical generation profile. In this work, we focus on radical flux as the defining kinetic parameter governing linear polymer formation within FRaP. Linear polymers are particularly challenging to form in this regime, as rapid polymer flow competes with reaction front dynamics.³⁷ Yet, linear polymers provide an ideal platform for mechanistic study owing to their

Received: February 13, 2026

Revised: April 17, 2026

Accepted: April 20, 2026



ACS Publications

© XXXX The Authors. Published by
American Chemical Society

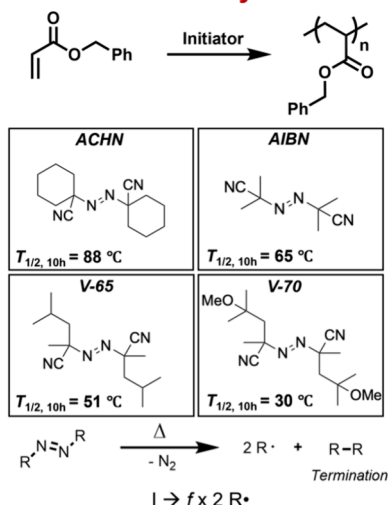
A

<https://doi.org/10.1021/acsmacrolett.6c00114>
ACS Macro Lett. XXXX, XXX, XXX–XXX

defined architecture and accessible characterization.^{38–40} By systematically varying initiator decomposition kinetics, we probe how transient radical flux dictates front stability,

propagation velocity, and molecular weight development. This radical-centric framework establishes new principles for controlling linear polymer synthesis under dynamic bulk

A Frontal Radical Polymerization of Linear Polymers



B Fitting $T(dt)$ for $[R\cdot]$ Calculations

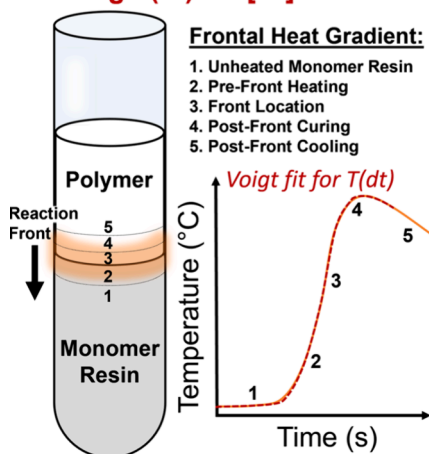


Figure 1. (A) FRaP scheme using varied initiators with 10 h half-life temperatures ($T_{1/2,10h}$). (B) Cartoon depiction of frontal polymerization heat gradient “zones” and thermal profile fit for temperature vs time.

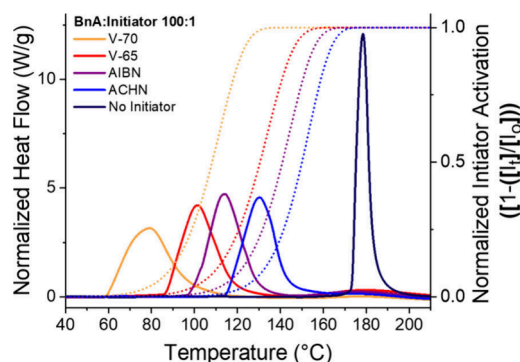


Figure 2. Differential scanning calorimetry results for all tested initiators and the autopolymerization of benzyl acrylate (BnA) overlaid with theoretical radical concentrations.

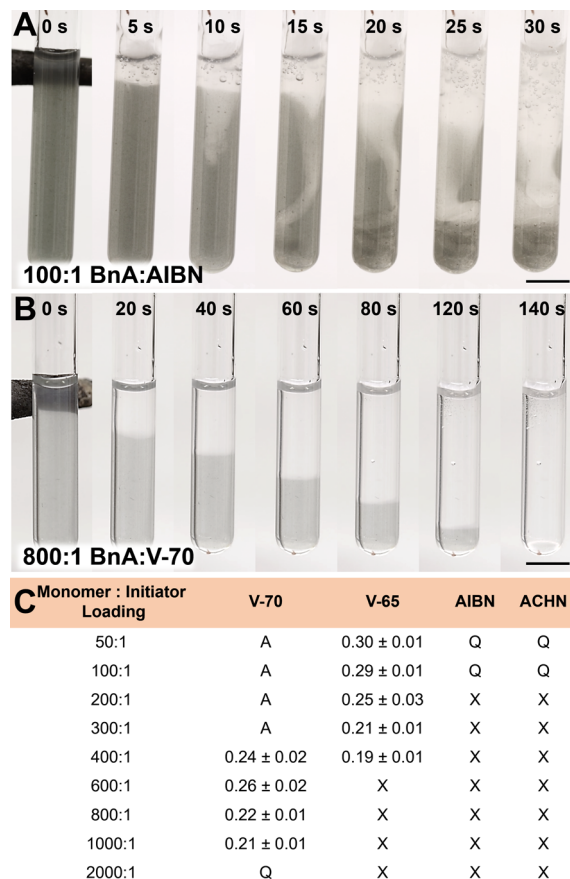


Figure 3. (A) Timelapse of 100:1 BnA:AIBN resin with thermochromic dye (reversible black to clear at 70 °C) showing polymer flow outcompeting frontal kinetics. (B) Timelapse of the FRaP of 800:1 BnA:V-70 loading (scale bars are 6 mm). (C) All tested initiator loadings for each azo initiator and front speed (mm/s), if thermally activated within 20 min, Q = quenched, and X = not tested.

conditions, offering a path toward harnessing FRaP for systems that have traditionally resisted frontally mediated processing.

To begin this work, we chose to vary azo-based initiators by the given 10 h half-life temperature ($T_{1/2,10h}$) while keeping the monomer species constant (Figure 1A) to understand how radical flux can be used to tune the front profile (Figure 1B). We chose benzyl acrylate (BnA) as it is a high boiling liquid, suppressing void formation from the elevated temperatures of FRaP and enabling facile mixing with initiators.⁴¹ Additionally, we chose to propagate frontal polymerization in the direction of gravity to avoid buoyancy-driven convection.^{3,42,43} Differential scanning calorimetry (DSC) thermograms of 100:1 BnA:initiator for 2,2'-azobis(4-methoxy-2,4-dimethylvaleronitrile) (V-70), 2,2'-azobis(2,4-dimethylvaleronitrile) (V-65), azobis(isobutyronitrile) (AIBN), 1,1'-azobis(cyclohexanecarbonitrile) (ACHN), and BnA autopolymerization (no initiator) were evaluated (Figure 2), showing the stark decrease of onset temperature to polymerization (exotherm from monomer consumption) with a trend following the $T_{1/2,10h}$ for each initiator. Using the heating rate of these experiments, we calculated the relative initiator consumption (Figures S1–S5,

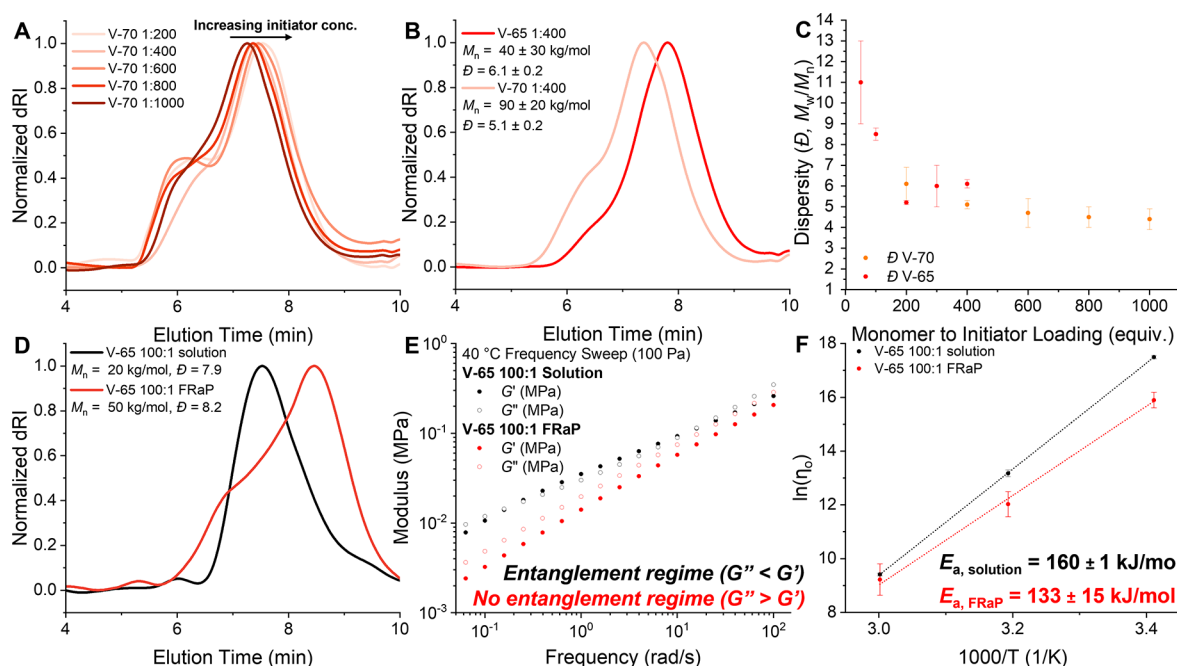


Figure 4. (A) Size-exclusion chromatography (SEC) of polybenzyl acrylate (pBnA) from varied V-70 loadings showing a large molecular weight shoulder and decrease of the main distribution with increasing initiator loading. (B) SEC of pBnA from V-70 and V-65 at 400:1 monomer:initiator showing a decrease in the molecular weight (M_n) with the higher temperature initiator (V-65). (C) Dispersity ($\bar{D}$) of pBnA from varied V-70 (orange) and V-65 (red) loadings showing the influence of initiator loading. (D) SEC of solution- and FRaP-derived pBnA at 100:1 BnA:V-65. (E) Frequency sweep at 40 °C of solution-derived (black) and FRaP-derived (red) pBnA showing only viscoelastic fluid behavior from lack of entanglements for FRaP-derived samples even at a larger M_n compared to solution-derived samples. (F) Arrhenius plots from creep-recovery experiments at 60, 40, and 20 °C with calculated activation energy for viscous flow (E_a).

Tables S1 and S2) showing the same trend and similar onset temperatures to the peak temperature from DSC. The peak temperature from DSC was 79, 102, 114, and 130 °C and maximum $[R^*]$ temperature was calculated to be 94, 135, 144, and 154 °C for V-70, V-65, AIBN, and ACHN, respectively.

Thus, we first attempted FRaP with AIBN, a commonly employed azo initiator, observing stable front propagation at high initiator-to-monomer loadings (i.e., 25:1 molar ratio of BnA:AIBN). Lower initiator loadings consistently resulted in front quenching from polymer flow dispersing local heat, making these conditions unsuitable for frontal polymerization (Figure 3A).

To improve efficiency and reduce initiator loading, V-70 was explored due to its lower onset temperature for radical formation. V-70 demonstrated improved performance, supporting stable and continuous front propagation at significantly reduced loadings, down to 1000:1 (Figure 3B). Monomer-to-initiator ratios of 400:1, 600:1, 800:1, and 1000:1 yielded front velocities (v_f) of 0.24 ± 0.02 , 0.26 ± 0.02 , 0.22 ± 0.01 , and 0.21 ± 0.01 mm/s, respectively, showcasing V-70's ability to sustain a front and outcompete the inherent polymer flow for these systems. Higher loadings of initiator also displayed frontal polymerization but would autopolymerize within 20 min, precluding such formulations from further studies.

Encouraged by these results, we then evaluated frontal polymerization using V-65. With a slightly higher onset temperature than V-70, V-65 enabled higher initiator loadings without autopolymerization. Monomer-to-initiator ratios of 50:1, 100:1, 200:1, 300:1, and 400:1 produced v_f of 0.30 ± 0.01 , 0.29 ± 0.01 , 0.25 ± 0.03 , 0.21 ± 0.01 , and 0.19 ± 0.01 mm/s, respectively (Figures 3C and S6–S27). Notably, linear and stable front propagation was only observed when v_f exceeded 0.2

mm/s for both initiators, suggesting a threshold for polymerization rate to outcompete polymer flow that interrupts the exothermic feedback loop required for self-sustaining propagation. To further investigate the influence of initiator thermal sensitivity, ACHN, possessing the highest onset temperature among the initiators studied, was tested. ACHN failed to support front propagation at all loadings, reinforcing the critical role of low-temperature radical generation in enabling and sustaining a propagating front.

Due to their linear architecture, these polymers are readily soluble and amenable to characterization techniques not available to thermosets. Polymer samples activated with V-65 and V-70 were analyzed for molecular weight and dispersity ($\bar{D}$; Figures 4A–C, S28–S31, and Tables S3 and S4). In both cases, decreasing the initiator loading led to a shift in size exclusion chromatography (SEC) peaks toward higher number-average molecular weights (M_n), a similar trend observed for solution state polymerization.⁴⁴ $\bar{D}$ values were large at all loadings but tended to decrease with lower initiator loadings. Additionally, glass transition temperature (T_g) values decreased with lower initiator loadings, attributed to plasticization from residual monomer and low-molecular-weight chains (Figures S32–S55).

For comparison, a solution-state V-65 sample was prepared at 100:1 BnA:V-65 and polymerized at 1 M in toluene at 50 °C for 3 days to ensure complete initiator consumption. The SEC trace of this control revealed a distribution skewed toward longer chains, in contrast to the purified FRaP samples, which—even at higher M_n —were skewed toward lower molecular weight distributions (Figure 4D). This divergence arises from fundamental kinetic and thermal transport considerations. In the solution-state polymerization, the low concentration allows radicals to persist longer before undergoing bimolecular

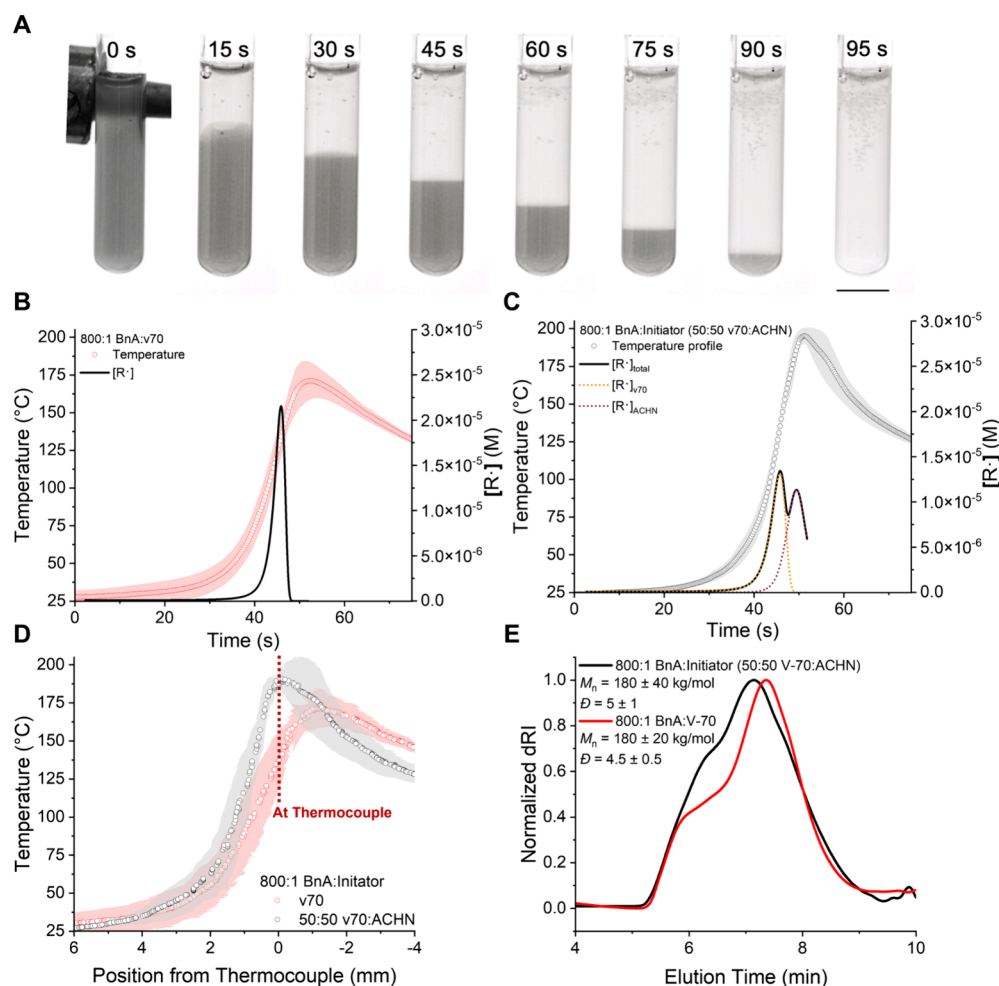


Figure 5. (A) Timelapse of 800:0.5:0.5 BnA:V-70:ACHN with a thermochromic dye for visualization (scale bar is 6 mm). (B) Thermal profile of 800:1 BnA:V-70 with calculated radical flux showing a monomodal radical decomposition profile. (C) Thermal profile of 800:0.5:0.5 BnA:V-70:ACHN with calculated radical flux showing a bimodal radical decomposition profile. (D) Temperature vs position from thermocouple and front position tracking. (E) Size-exclusion chromatography showing both V-70 and 50:50 ACHN:V-70 FRAp-derived polymers showing identical number-average molecular weight (M_n) and dispersity ($\bar{D}$).

termination. As propagation outpaces termination, a fraction of polymers achieve large chain lengths, skewing toward higher molecular weights despite modest M_n . In FRAp, ahead of the propagating front, limited radical availability favors chain growth and the formation of polymers with higher M_n . As the front passes and local temperatures rise sharply, steep thermal gradients generate high radical fluxes from small-molecule initiators, promoting rapid initiation and concomitant termination of short chains and skewing the molecular weight distribution toward lower M_n .

This asymmetry in distribution has critical rheological consequences. Despite the higher M_n of the FRAp-derived polymers, the dominance of low-molecular-weight species effectively disrupts entanglement formation, yielding materials with rheological behavior more reminiscent of unentangled melts.^{7,45,46} By contrast, the solution-state controls, although possessing lower M_n overall, demonstrated skewness toward viscoelastic behavior of an entangled melt.

This distinction is directly supported by small-amplitude oscillatory shear experiments. The solution-state polymers displayed a transition from viscoelastic solid ($G' > G''$) to viscoelastic fluid behavior, indicative of network-like entanglement effects.⁴⁷ Conversely, the FRAp polymers, lacking

sufficient chain entanglement due to their low molecular weight skewness, exhibited purely viscoelastic fluid behavior ($G' < G''$ at all frequencies), consistent with an unentangled linear melt (Figure 4E). Furthermore, the creep-recovery experiments revealed a higher apparent activation energy for viscous flow (higher heat sensitivity) for the entangled solution-derived samples, underscoring the greater energetic barrier associated with reptation and disentanglement of long-chain populations (Figure 4F). Together, these findings highlight that not only average molecular weight, but molecular weight distribution skewness, dictates the rheological landscape of linear FRAp-derived polymers.^{48–52}

Building on our understanding of radical flux in FRAp, we hypothesized that combining a low-temperature initiator (V-70) with a high-temperature initiator (ACHN) could yield sequential radical bursts during propagation. Thus, this cooperative initiation could enable lower initiator loadings than are accessible with either initiator alone. In this dual-initiator design, V-70 would decompose at low temperature, gently preheating the resin ahead of the front and activating ACHN for a subsequent, high-temperature radical burst. We selected a 1:1 ratio of V-70 and ACHN at an overall initiator loading of 800:1 (monomer:initiator molar ratio), as V-70 alone

at 1600:1 cannot sustain a self-propagating front—requiring the secondary ACHN activation to maintain the reaction cascade.

Excitingly, this dual-initiator formulation supported stable frontal polymerization (Figures 5A and S56–S60 and Tables S5–S8) with a v_f of 0.18 ± 0.01 mm/s and a maximum front temperature of 195 ± 4 °C, notably higher than the front temperature (174 ± 8 °C) and slower v_f observed for the single-initiator (800:1 BnA:V-70) control. To quantify these behaviors, temperature–time data were fit using a Voigt model for $T(t)$, capturing both the rise and initial decay of thermal intensity. These fits revealed a single radical burst for the V-70-only system, consistent with its unimodal decomposition profile, while the dual-initiator system exhibited two distinct bursts corresponding to sequential activation of V-70 then ACHN (Figure 5B,C). This dual flux not only reinforced our mechanistic hypothesis but also underscores how initiator pairing can be used to program radical dynamics and front energetics in reactive manufacturing of vinyl polymers. By tracking the front position relative to the embedded thermocouple, the breadth of the thermal gradient could be directly visualized both ahead of and behind the visible reaction front (Figures 5D and S61). The standalone V-70 system exhibited a broad thermal profile, consistent with diffuse preheating before front arrival and sluggish cooling after front passage—reflecting gradual radical release and diffusion-mediated propagation. In contrast, the 1:1 V-70:ACHN formulation displayed a much sharper heat gradient, indicative of more compact energy release and efficient confinement of reaction exothermicity within the advancing front. Finally, both monoliths were analyzed by DSC, NMR, and SEC, yielding identical conversion, M_n , and $\bar{D}$ (Figures 5E, S62, and S63). However, the chain-length distribution was not equivalent, showing a pronounced high molecular weight shoulder for the mixed initiator system, which we attributed to the sequential radical activation of the mixed system limiting termination. This trend can be inverted or completely altered depending on loading and initiator type (Figures S64–S69).

Understanding and directing radical flux represents a pivotal step toward precision in frontal polymerization. By resolving how the timing and magnitude of radical generation govern front stability, conversion kinetics, and chain growth, we uncover a framework for programming reactivity through initiator design and thermal transport. These insights open new opportunities to access and interrogate un-cross-linked polymers in frontal systems that have historically been inaccessible to microstructural analysis.

■ ASSOCIATED CONTENT

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsmacrolett.6c00114>.

Experimental details (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Jacob J. Lessard – Department of Chemistry, University of Utah, Salt Lake City, Utah 84112, United States of America;

orcid.org/0000-0003-2962-6472; Email: jacob.lessard@utah.edu

Authors

Darya A. Ivannikava – Department of Chemistry, University of Utah, Salt Lake City, Utah 84112, United States of America

Audrey N. McMullin – Department of Chemistry, University of Utah, Salt Lake City, Utah 84112, United States of America

Kevin A. Stewart – Department of Chemistry, University of Utah, Salt Lake City, Utah 84112, United States of America

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acsmacrolett.6c00114>

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was sponsored by faculty startup funds provided by the Department of Chemistry at the University of Utah. The authors would also thank Michayla L. Hartenfels for administrative support.

■ REFERENCES

- (1) Suslick, B. A.; Hemmer, J.; Groce, B. R.; Stawiasz, K. J.; Geubelle, P. H.; Malucelli, G.; Mariani, A.; Moore, J. S.; Pojman, J. A.; Sottos, N. R. Frontal Polymerizations: From Chemical Perspectives to Macroscopic Properties and Applications. *Chem. Rev.* **2023**, *123*, 3237–3298.
- (2) Robertson, I. D.; Yourdkhani, M.; Centellas, P. J.; Aw, J. E.; Ivanoff, D. G.; Goli, E.; Lloyd, E. M.; Dean, L. M.; Sottos, N. R.; Geubelle, P. H.; et al. Rapid energy-efficient manufacturing of polymers and composites via frontal polymerization. *Nature* **2018**, *557*, 223–227.
- (3) Stewart, K. A.; Ivannikava, D. A.; Massouh, C. M.; Lessard, J. J. Molecular Weight Control in Frontal Ring-Opening Metathesis Polymerization. *Angew. Chem., Int. Ed.* **2025**, *64*, No. e202510071.
- (4) Bomze, D.; Knaack, P.; Koch, T.; Jin, H.; Liska, R. Radical induced cationic frontal polymerization as a versatile tool for epoxy curing and composite production. *Polym. Sci. A Polym. Chem.* **2016**, *54*, 3751–3759.
- (5) Maafa, I. M. Inhibition of Free Radical Polymerization: A Review. *Polymers* **2023**, *15*, 488.
- (6) Dakshinamoorthy, D.; Khopkar, A. R.; Louvar, J. F.; Ranade, V. V. CFD simulations to study shortstopping runaway reactions in a stirred vessel. *J. Loss Prev. Process Ind.* **2004**, *17*, 355–364.
- (7) Kamiyama, Y.; Tamate, R.; Hiroi, T.; Samitsu, S.; Fujii, K.; Ueki, T. Highly stretchable and self-healable polymer gels from physical entanglements of ultrahigh-molecular weight polymers. *Sci. Adv.* **2022**, *8*, No. eadd0226.
- (8) Norrish, R. G. W.; Brookman, E. F. The Mechanism of Polymerization Reactions. I. The Polymerization of Styrene and Methyl Methacrylate. *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences* **1939**, *171*, 147–171.
- (9) Norrish, R. G. W.; Smith, R. R. Catalysed Polymerization of Methyl Methacrylate in the Liquid Phase. *Nature* **1942**, *150*, 336–337.
- (10) Trommsdorff, V. E.; Köhle, H.; Lagally, P. Zur polymerisation des methacrylsäuremethylesters. *Makromol. Chem.* **1948**, *1*, 169–198.
- (11) Stewart, K. A.; Lombardi, F. J.; Cao, S. D.; Massouh, C. M.; Lessard, J. J. Engineering Free Volume within Frontal Ring-Opening Metathesis Polymerization via Pendant Plasticization. *Adv. Mater.* **2026**, *38*, No. e19676.
- (12) Pojman, J. A.; Ilyashenko, V. M.; Khan, A. M. Free-radical frontal polymerization: self-propagating thermal reaction waves. *Journal of the Chemical Society, Faraday Transactions* **1996**, *92*, 2825–2837.
- (13) Arretche, I.; Lessard, J. J.; Kaur, P.; Mills, M. G.; Kim, A. J.; Liu, H. J.; Cooper, J. C.; Ewoldt, R. H.; Moore, J. S.; Tawfick, S. Active Learning Guided Optimization of Frontal Ring-Opening Metathesis

Polymerization via Alkylidene Modification. *ACS Macro Lett.* **2025**, *14*, 525–531.

(14) Cramblitt, A. C.; Paul, J. E.; Price, T. C.; Kaur, P.; Moore, J. S.; Sottos, N. R.; Lessard, J. J. Chemical Activation of Frontal Ring-Opening Metathesis Polymerization. *ACS Macro Lett.* **2025**, *14*, 1182–1188.

(15) Wang, H. M.; Suslick, B. A.; Lessard, J. J.; Yu, C. H.; Saengow, C.; Ewoldt, R. H.; Moore, J. S. Frontal Polymerization of Stereoisomers: Distinct Microstructures and Properties in Endo- and Exo-Dicyclopentadiene Thermosets. *Macromolecules* **2024**, *57*, 142–149.

(16) Lessard, J. J.; Mejia, E. B.; Kim, A. J.; Zhang, Z.; Berkey, M. G.; Medina-Barreto, Z. S.; Ewoldt, R. H.; Sottos, N. R.; Moore, J. S. Unraveling Reactivity Differences: Room-Temperature Ring-Opening Metathesis Polymerization (ROMP) versus Frontal ROMP. *J. Am. Chem. Soc.* **2024**, *146*, 7216–7221.

(17) Bistri, D.; Arretche, I.; Lessard, J. J.; Zakowrotny, M.; Vyas, S.; Rongy, L.; Gómez-Bombarelli, R.; Moore, J. S.; Geubelle, P. A Mechanism-Based Reaction–Diffusion Model for Accelerated Discovery of Thermoset Resins Frontally Polymerized by Olefin Metathesis. *J. Am. Chem. Soc.* **2024**, *146*, 21877–21888.

(18) Alzate-Sanchez, D. M.; Yu, C. H.; Lessard, J. J.; Paul, J. E.; Sottos, N. R.; Moore, J. S. Rapid Controlled Synthesis of Large Polymers by Frontal Ring-Opening Metathesis Polymerization. *Macromolecules* **2023**, *56*, 1527–1533.

(19) Lessard, J. J.; Kaur, P.; Paul, J. E.; Chang, K. M.; Sottos, N. R.; Moore, J. S. Switching Frontal Polymerization Mechanisms: FROMP and FRaP. *ACS Macro Lett.* **2022**, *11*, 1097–1101.

(20) Paul, J. E.; Gao, Y.; Go, Y. K.; Rodriguez Koett, L. E.; Sharma, A.; Chen, M.; Lessard, J. J.; Topkaya, T.; Leal, C.; Moore, J. S.; et al. Controlled patterning of crystalline domains by frontal polymerization. *Nature* **2024**, *634*, 85–90.

(21) Ziaee, M.; Johnson, J. W.; Yourdkhani, M. 3D Printing of Short-Carbon-Fiber-Reinforced Thermoset Polymer Composites via Frontal Polymerization. *ACS Appl. Mater. Interfaces* **2022**, *14*, 16694–16702.

(22) Nuvoli, D.; Alzari, V.; Pojman, J. A.; Sanna, V.; Ruii, A.; Sanna, D.; Malucelli, G.; Mariani, A. Synthesis and Characterization of Functionally Gradient Materials Obtained by Frontal Polymerization. *ACS Appl. Mater. Interfaces* **2015**, *7*, 3600–3606.

(23) Pojman, J. A. 4.38 - Frontal Polymerization. In *Polymer Science: A Comprehensive Reference*; Matyjaszewski, K., Möller, M., Eds.; Elsevier, 2012; pp 957–980.

(24) Pojman, J. A. Frontal Polymerization. In *Polymer Science: A Comprehensive Reference*; Matyjaszewski, K., Möller, M., Eds.; Elsevier, 2012; pp 957–980.

(25) Pujari, N.; Inamdar, S.; Ponrathnam, S.; Kulkarni, B. Water-Triggered Frontal Polymerization. *Macromol. Rapid Commun.* **2007**, *28*, 109–115.

(26) Nason, C.; Roper, T.; Hoyle, C.; Pojman, J. A. UV-Induced Frontal Polymerization of Multifunctional (Meth)acrylates. *Macromolecules* **2005**, *38* (13), 5506–5512.

(27) Pojman, J. A.; Popwell, S.; Fortenberry, D. I.; Volpert, V. A.; Volpert, V. A. Nonlinear Dynamics in Frontal Polymerization. In *Nonlinear Dynamics in Polymeric Systems*, ACS Symposium Series; American Chemical Society, 2003; Vol. 869, pp 106–120.

(28) Fiori, S.; Mariani, A.; Ricco, L.; Russo, S. Interpenetrating polydicyclopentadiene/polyacrylate networks obtained by simultaneous non-interfering frontal polymerization. *e-Polym.* **2002**, *2* (1), na DOI: 10.1515/epoly.2002.2.1.404.

(29) Mariani, A.; Fiori, S.; Chekanov, Y.; Pojman, J. A. Frontal Ring-Opening Metathesis Polymerization of Dicyclopentadiene. *Macromolecules* **2001**, *34*, 6539–6541.

(30) Chekanov, Y. A.; Pojman, J. A. Preparation of functionally gradient materials via frontal polymerization. *J. Appl. Polym. Sci.* **2000**, *78*, 2398–2404.

(31) Masere, J.; Stewart, F.; Meehan, T.; Pojman, J. A. Period-doubling behavior in frontal polymerization of multifunctional acrylates. *Chaos* **1999**, *9*, 315–322.

(32) Chechilo, N.; Enikolopyan, N. Structure of the polymerization wave front and propagation mechanism of the polymerization reaction. *Dokl. Phys. Chem.* **1974**, *214*, 174–176.

(33) Chechilo, N.; Khvilivitskii, R.; Enikolopyan, N. On the phenomenon of polymerization reaction spreading. In *Dokl. Akad. Nauk SSSR* **1972**, *204*, 1180–1181.

(34) Odian, G. *Principles of Polymerization*, 4th ed.; John Wiley & Sons, Inc, 2004.

(35) Bamford, C. H.; Tipper, C. F. H. *Comprehensive Chemical Kinetics; Free Radical Polymerization*; Elsevier Scientific Publishing Company, 1976.

(36) Goldfeder, P. M.; Volpert, V. A.; Ilyashenko, V. M.; Khan, A. M.; Pojman, J. A.; Solovoyov, S. E. Mathematical Modeling of Free-Radical Polymerization Fronts. *J. Phys. Chem. B* **1997**, *101*, 3474–3482.

(37) Pojman, J. A.; Willis, J.; Fortenberry, D.; Ilyashenko, V.; Khan, A. M. Factors affecting propagating fronts of addition polymerization: Velocity, front curvature, temperature profile, conversion, and molecular weight distribution. *J. Polym. Sci. A Polym. Chem.* **1995**, *33*, 643–652.

(38) Volpert, V. A.; Volpert, V. A.; Ilyashenko, V. M.; Pojman, J. A. Frontal polymerization in a porous medium. *Chem. Eng. Sci.* **1998**, *53*, 1655–1665.

(39) Lemaigre, L.; Budroni, M. A.; Riolfo, L. A.; Grosfils, P.; De Wit, A. Asymmetric Rayleigh–Taylor and double-diffusive fingers in reactive systems. *Phys. Fluids* **2013**, *25* (1), na DOI: 10.1063/1.4774321.

(40) Nagy, I. P.; Pojman, J. A. Suppressing Convective Instabilities in Propagating Fronts by Tube Rotation. *J. Phys. Chem.* **1996**, *100*, 3299–3304.

(41) Pojman, J. A.; Gunn, G.; Patterson, C.; Owens, J.; Simmons, C. Frontal Dispersion Polymerization. *J. Phys. Chem. B* **1998**, *102*, 3927–3929.

(42) Bidali, S.; Ducrot, A.; Mariani, A.; Rustici, M. Self-ignition of polymerization fronts with convection: the ‘rainstorm effect’. *e-Polymers* **2005**, *5* (1), na.

(43) McCaughey, B.; Pojman, J. A.; Simmons, C.; Volpert, V. A. The effect of convection on a propagating front with a liquid product: Comparison of theory and experiments. *Chaos* **1998**, *8*, 520–529.

(44) Stewart, K. A.; Le, T. H.; Wei, N. B.; Cantor, A. J.; Rynk, J. F.; Sumerlin, B. S. Biobased Polystyrene Vitrimers for Thermoset Circularity. *ACS Appl. Polym. Mater.* **2024**, *6*, 13034–13041.

(45) Fox, T. G., Jr.; Flory, P. J. Second-Order Transition Temperatures and Related Properties of Polystyrene. I. Influence of Molecular Weight. *J. Appl. Phys.* **1950**, *21*, 581–591.

(46) Flory, P. J. *Principles Of Polymer Chemistry*; Cornell University Press, 1953.

(47) Marquez, J. D.; Stewart, K. A.; Stevens, K. C.; Ryder, B. J.; Le, T. H.; Wei, N. B.; Choi, W. J.; Huang, Y.; Sumerlin, B. S.; Evans, A. M. Tuning Entanglement Molecular Weights with Ultrahigh-Molecular-Weight Polystyrenics. *J. Am. Chem. Soc.* **2025**, *147*, 29223–29231.

(48) Gentekos, D. T.; Sifri, R. J.; Fors, B. P. Controlling polymer properties through the shape of the molecular-weight distribution. *Nature Reviews Materials* **2019**, *4*, 761–774.

(49) Gentekos, D. T.; Dupuis, L. N.; Fors, B. P. Beyond Dispersity: Deterministic Control of Polymer Molecular Weight Distribution. *J. Am. Chem. Soc.* **2016**, *138*, 1848–1851.

(50) Whitfield, R.; Truong, N. P.; Messmer, D.; Parkatzidis, K.; Rolland, M.; Anastasaki, A. Tailoring polymer dispersity and shape of molecular weight distributions: methods and applications. *Chem. Sci.* **2019**, *10*, 8724–8734.

(51) Whitfield, R.; Truong, N. P.; Anastasaki, A. Precise Control of Both Dispersity and Molecular Weight Distribution Shape by Polymer Blending. *Angew. Chem., Int. Ed.* **2021**, *60*, 19383–19388.

(52) Rosenbloom, S. I.; Hsu, J. H.; Fors, B. P. Controlling the shape of the molecular weight distribution for tailored tensile and rheological properties in thermoplastics and thermoplastic elastomers. *J. Polym. Sci.* **2022**, *60*, 1291–1299.